

Name : _____ Signature: _____

July 2025

S5 Mid term 2 Examinations

CHEMISTRY theory

Paper 1

2 hours

Uganda Advanced Certificate of Education

CHEMISTRY

(Principal subject)

Paper 1

2 hours

INSTRUCTIONS TO CANDIDATES

- This paper consists of ten (10) examination items; all items are compulsory and carry equal scores.
- Responses are to be written only in the spaces provided.
- A Periodic Table is provided, refer to it whenever needed.

Section A**Respond to all items in this section****Item 1**

In Nakaseke, Uganda, a local water treatment initiative aims to address contamination in groundwater caused by heavy metal runoff from agricultural activities. To mitigate environmental impact, the community is exploring the use of a metal-based filtration system to remove toxic ions from water sources. Two elements, X (neutron number 4 and atomic mass 7) and Y (atomic mass 39 and neutron number 20), are being considered for the filtration material due to their potential to bind with contaminants. However, the chosen element must have low reactivity to ensure the filtration system remains stable, durable, and does not introduce additional chemical reactions that could harm the environment.

Tasks

Using your knowledge of Chemistry,

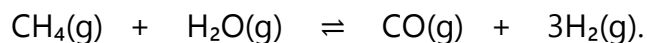
- a) Compare the reactivity of elements X and Y, explaining how their electron configurations influence their chemical reactivity.

- b) Advise the Nakaseke water treatment initiative on which element (X or Y) is the better choice for the filtration system, justifying your recommendation based on lower reactivity to minimize environmental impact.

- c) Briefly explain how selecting the less reactive element supports the environmental goals of the water treatment project in Nakaseke.

Item 2

EcoGas and Green Energy Ltd are two gas companies that sell water gas (a mixture of carbon monoxide and hydrogen). They prepare this gas by the chemical process (natural gas reformation) below:



EcoGas uses a scaled down reaction mixture that is 1200 liters methane and 1500 liters steam; Green Energy uses a mixture that is 1350 liters methane and an equal volume of steam at the high conditions of temperature and pressure required for the operation.

Tasks:

As a student of Chemistry approached by a company hoping to make the economical purchase from these two companies that sell their gas at the same price per kilogram and claim that replacing charcoal with their gas as a fuel is better for environmental protection.

- a) Determine which of the products from the two companies has the less impurity by percentage.

- b) Advise the purchasing company on the better choice. Give reasons for your advise.

c) Evaluate the products from these companies as compared to charcoal.

Item 3

Technetium-99m is the most commonly used radioisotope for medical imaging. It's widely employed in diagnostic procedures like SPECT scans due to its ideal half-life of about 6 hours and its ability to emit gamma rays suitable for imaging organs and tissues. Its working activity is 1110 to 370 MBq below which gamma ray cameras fail to capture useful organ or tissue images. A nurse at Mulago cancer ward is faced with a dilemma; she is to administer a dose of the drug with technetium - 99m to a patient when its activity reads 880 MBq and the imaging technician is to report 15.5 hours later.

Tasks

As a student of Chemistry, study the situation and,

- a) Calculate the activity of the radioisotope at the time the imaging technician reports back to hospital.
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b) Advise the nurse and give a reason for your advise.

c) In addition to the advice in (b) above advise her on how she should safeguard her health at her job.

Item 4

In Kotido District, Uganda, the Kaabong Community Police Department has enlisted the help of a forensics intern, Amina Lokoru, to recreate a crime scene involving a metallic alloy used in a suspicious device. The alloy, suspected to be a reactive but relatively stable material, was crafted from elements in Group 1A (alkali metals) and Group 2A (alkaline earth metals). Amina needs your expertise in understanding the variation in atomic radii down a group in the periodic table to select appropriate elements for reconstructing the alloy. Your task is to guide her in choosing elements from Group 1A and Group 2A based on their atomic radii and explain how these choices impact the alloy's reactivity and stability for forensic analysis.

Tasks:

Amina is considering elements from Group 1A (Li, Na, K) and Group 2A (Be, Mg, Sr and Ca) to recreate the alloy used in the crime scene device. She needs your help to:

- Select one element from Group 1A and one from Group 2A to form a reactive but relatively stable alloy. Explain how the variation in atomic radii

down each group influences your choice, and predict how the atomic radii affect the alloy's reactivity and stability for forensic reconstruction.

[illegible]

b) Compare the reactivity of your chosen Group 1A element with another Group 1A element further down the group, explaining how the difference in atomic radii impacts the alloy's properties.

c) Provide a brief explanation of why atomic radii increase down a group.

Item 5

In a beverage production facility in Kampala, Uganda, a company manufactures both carbonated soda and alcoholic beverages. The production team observes two key practices: (1) glass soda bottles are never filled to the brim, leaving a small headspace at the top, and (2) ethanol is diluted with water to produce consumable liquor, such as a popular local spirit. The facility manager wants to ensure that new trainees understand the chemical and physical principles behind these practices to maintain product quality and safety.

Tasks

- a) Use your knowledge of bonding and structure to educate the new trainees about,

- (i) why glass soda bottles are not filled to the brim, and the soda put in a deep freezer.

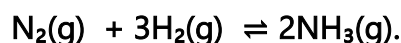
- (ii) why ethanol can be diluted with water to form a consumable liquor, addressing the interactions between the molecules.

- b) A health campaign to sensitize consumers about micro plastics in their food and drink is growing on radio talk shows and on social media. A part of the message is that government should ban the packing of alcohol in plastic

bottles. The scientific claims behind this proposal is that alcohol can slowly dissolve plastics including those based on hydrocarbons .
Evaluate these claims on a basis of bonding types.

Item 6

In a chemical manufacturing plant in Jinja, Uganda, engineers are optimizing the production of ammonia (NH_3) for fertilizer using the Haber-Bosch process, which involves the reaction of nitrogen (N_2) and hydrogen (H_2) gases:



The reaction is **exothermic** and requires a catalyst to be economically viable. The plant manager needs trainees to understand the energy changes involved to improve efficiency and reduce energy costs, particularly by analysing the reaction's energy profile and the role of activation energy in the reaction mechanism.

Task:

As a student of Chemistry:

- Synthesize an energy Profile for the catalyzed and uncatalyzed Haber-Bosch reaction, that can be used to explain to the trainees how efficiency can be improved and energy costs cut down.

- b) Evaluate the role of activation energy using the energy profiles above, explain how the activation energy influences the reaction mechanism and rate of ammonia production in both the catalyzed and uncatalyzed pathways.

- c) Discuss how the catalyst affects the activation energy and why this is critical for the industrial process.

Item 7

In a regional science contest held in Gulu, Uganda, secondary school learners are tasked with designing a solar energy storage system to provide reliable power for a rural community. The challenge involves selecting materials for two key components: an electrolyte for a battery to store energy, requiring a compound that conducts ions effectively and remains stable, and a conductive wire for efficient energy transfer, needing a material with excellent electrical conductivity and durability. The learners propose a choice between sodium chloride or aluminium bromide for the electrolyte and copper or magnesium for the wire. To win the contest, they must analyse the chemical principles behind these material choices to demonstrate their suitability for the solar energy system.

Task:

As a student of Chemistry

- a) Make a choice for the learners, and,

- b) Help them understand the chemical principles behind the suitability of the materials you have proposed in (a) above.

[illegible]

Item 8

In Gulu, Uganda, a community develops a water purification system to tackle groundwater contamination from agricultural runoff, using ammonia (NH_3) and carbon dioxide (CO_2) to form ammonium bicarbonate (NH_4HCO_3) as a coagulant. The team must evaluate NH_3 and CO_2 's molecular properties—polarity, bonding, and shapes—to

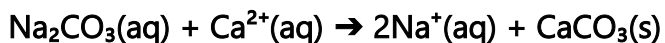
ensure effective and safe purification. The system requires compounds that support coagulation while prioritizing environmental safety.

Task : provide a brief write up explaining how ammonia and carbon dioxide's molecular characteristics enable NH_4HCO_3 formation, ensuring efficient impurity removal for Gulu's clean water supply.

[illegible]

Item 9

A water engineering firm uses anhydrous sodium carbonate that comes in 12kg bags each 65000 UqX, to remove calcium ions from hard water by the process:



A procurement officer has been tasked to estimate the cost of sodium carbonate that will be required to complete a job of softening 550,000 liter of a hard water containing 35.8mg of calcium ions per 200 cm³.

The procurement officer has sought after your help.

Task

- a) Showing a clear working, help the procurement officer estimate the number of bags of anhydrous sodium carbonate the job will require. (1 mg = 10^{-3} g; Na = 23, C = 12, O = 16 and Ca = 40)

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- b) If the National Drug Authority (NDA) and Uganda National Bureau of Standards (UNBS) recommend drinkable water to not contain more than 50mg per liter of sodium, determine whether the process of water treatment is lawful or not.

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1. The mass spectrum below was presented in a class plenary session by the teacher to your study group:



END